

Creating Building Datasets for CityBES

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1. Overview

The City Building Energy Saver (CityBES) is a web-based tool to support city-scale building energy efficiency. It is publicly available at <https://citybes.lbl.gov/>. Figure 1 shows the main screenshot of CityBES. CityBES is built upon Cesium, an open-source JavaScript library for world-class 3D globes and maps.

CityBES allows new building datasets for a city to be imported and used for building energy retrofit analysis, building energy benchmarking, and building performance visualization. It supports building datasets in CityGML and GeoJSON formats. The dataset includes two levels of building data: the minimally required, and the good-to-have optional data. This document describes the data requirements and approaches to creating a dataset for CityBES.

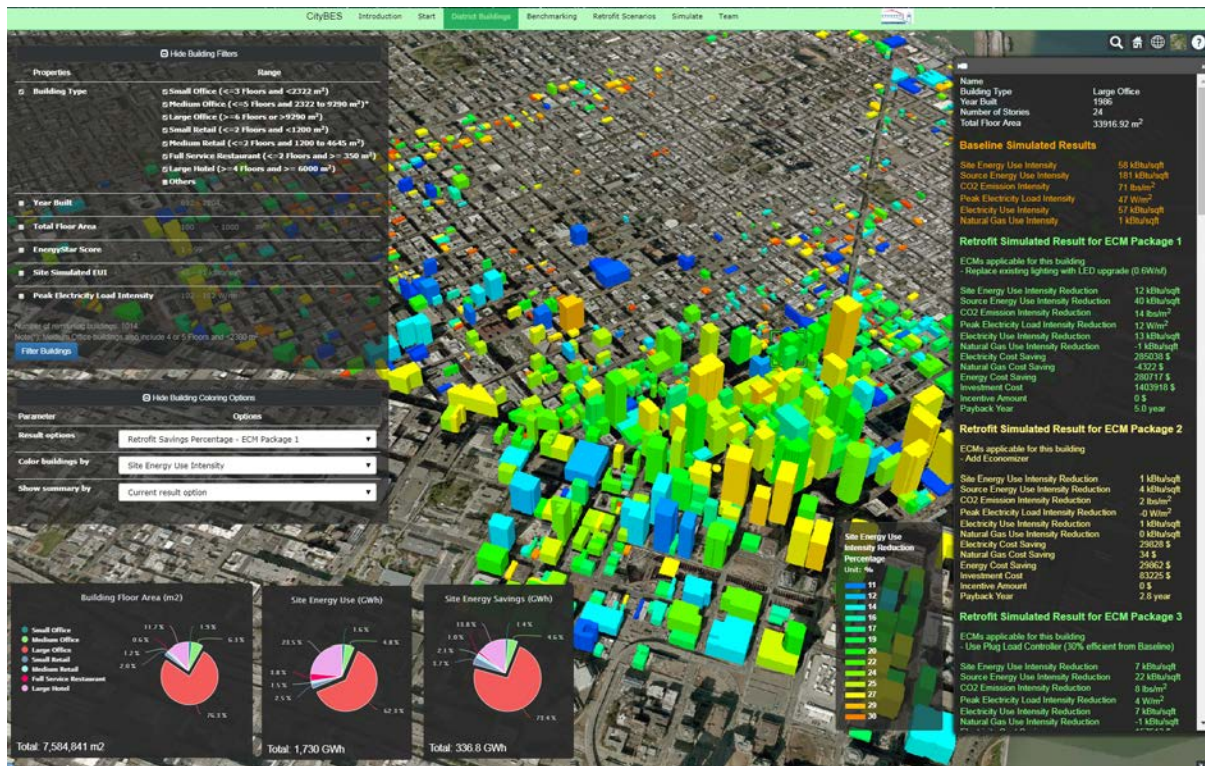


Figure 1. Screenshot of CityBES

2. Required Data for the CityBES Dataset

The required data for each building includes the building's geometry and its basic properties.

2.1. Building Geometry

CityBES requires the **building footprint** to create the geometry of the building model. Currently, two data formats are supported by CityBES: [GeoJSON](#) and [CityGML](#). If you are not familiar with GeoJSON or CityGML format, you can prepare the geometry data in the [Shapefile format](#) or [ESRI GDB format](#), which are typically available at cities' open data portals. Table 1 shows the example web portals for San Francisco, Chicago and New York City.

Table 1. Examples of building footprint in the shapefile format

City Name	Web portal
San Francisco	https://datasf.org/opendata/
Chicago	https://data.cityofchicago.org/
New York City	https://opendata.cityofnewyork.us/

You can use QGIS (<http://www.qgis.org/en/site/>) to visualize the GeoJSON, shapefile, and ESRI GDB files. The shapefile is usually distributed as a ZIP file. When you extract the file, you will see multiple files with the same name but different extensions. The file with ".shp" extension can be loaded into QGIS.

We recommend a maximum of about 25,000 buildings in each building dataset. If you wish to create datasets to cover the full city, you must split the data by planning districts or other similarly sized subdivisions. See Section 6 for how to do this. If you are only interested in a specific subset of buildings, you will need to prepare a shapefile with only the target buildings. Figure 2 shows the Chicago building footprint in QGIS. You can use the selection button to pick the target buildings as shown in Figure 3. Right click the dataset in the "Layers Panel" and select "Save as..." (Figure 4). You will see a pop window like Figure 5. For the "Format", select "GeoJSON". Choose a folder to save the file. Remember to check the "Save only selected features". Otherwise, it will save all the buildings rather than the selected buildings. Now you have a GeoJSON file with only the target buildings.

For Coordinate Reference System (CRS), Table 2 lists the CRS supported in CityBES. Project your datasets to one of these CRS (choose the CRS with the most similar climate). In Figure 4, click the CRS change button to use one of the CRS listed in Table 2.

Table 2. CRS currently supported by CityBES

CRS Name	Reference City
EPSG:2227	San Francisco, CA
EPSG:2229	Los Angeles, CA
EPSG:32118	New York City, NY
EPSG:2992	Portland, OR
EPSG:2253	Chicago, IL

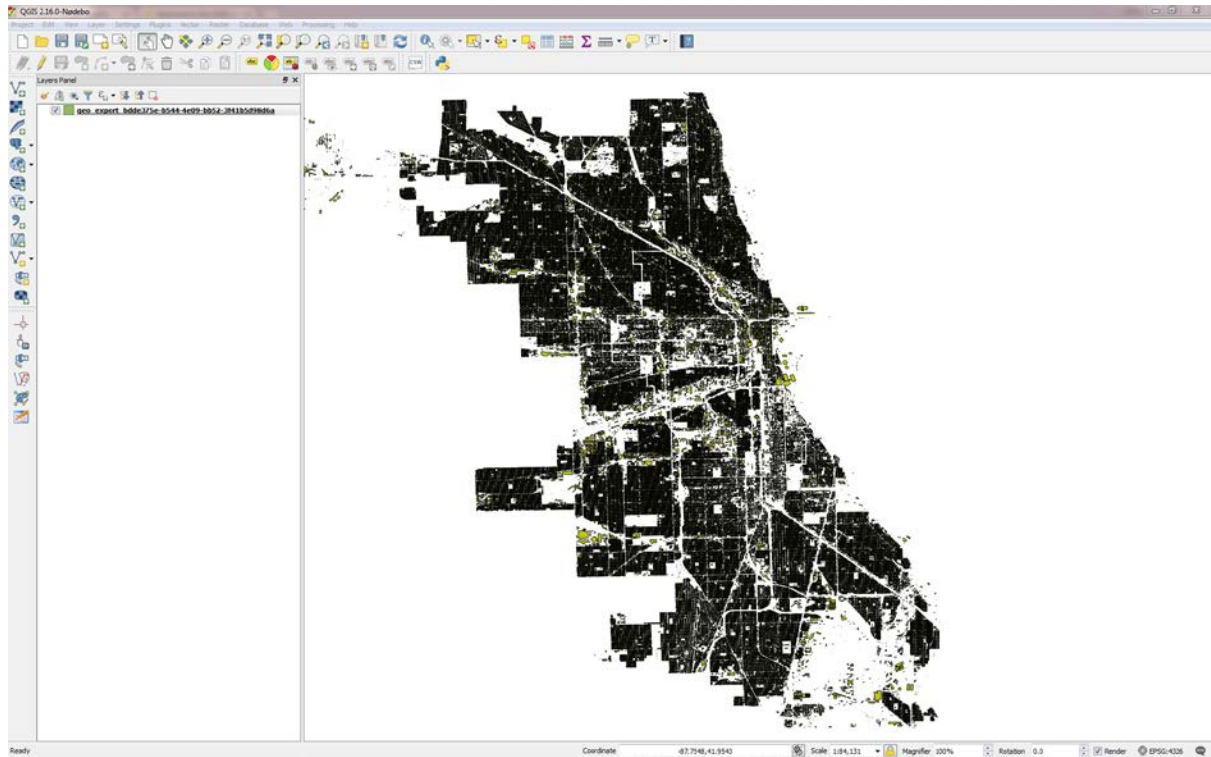


Figure 2. Visualization of Chicago building footprint in QGIS

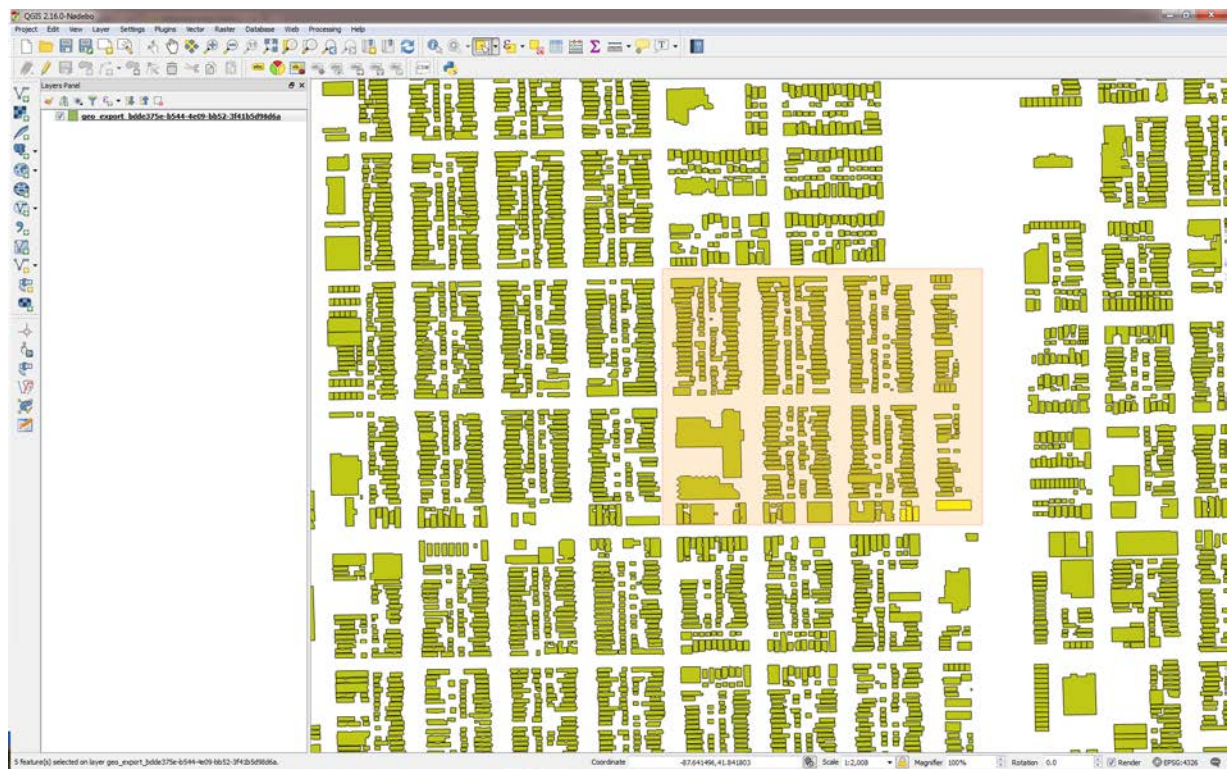


Figure 3. Select target buildings in QGIS

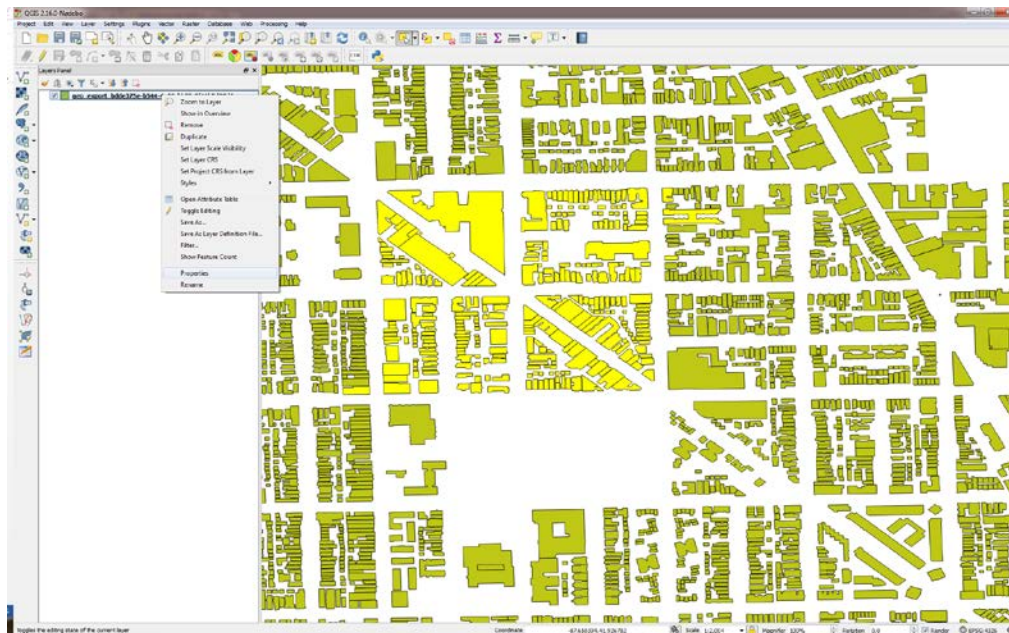


Figure 4. Save the target buildings as a new shapefile in QGIS

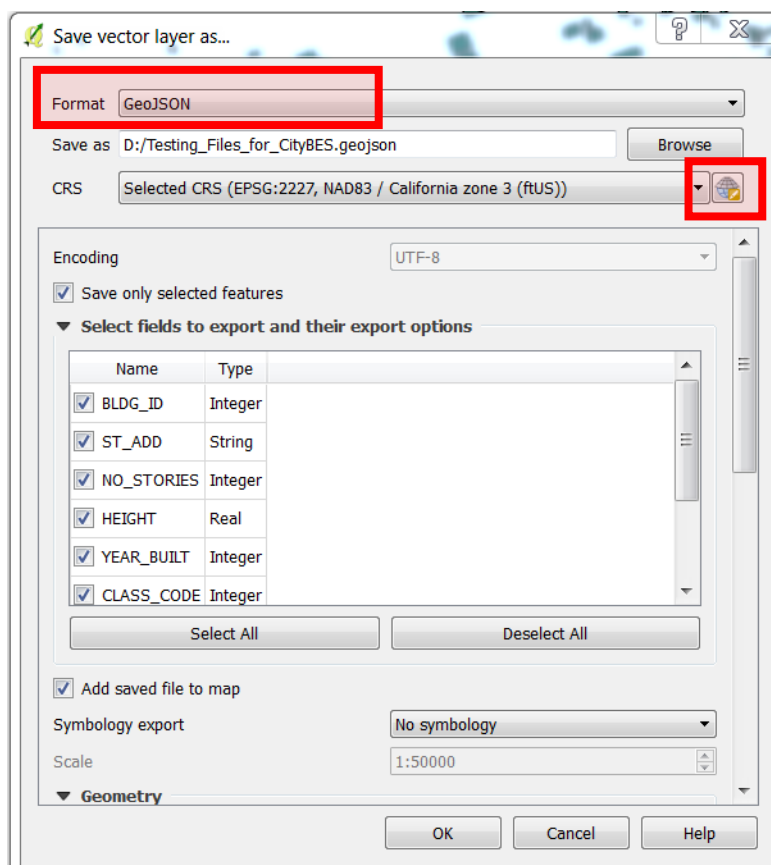


Figure 5. Save the target buildings as a new GeoJSON in QGIS

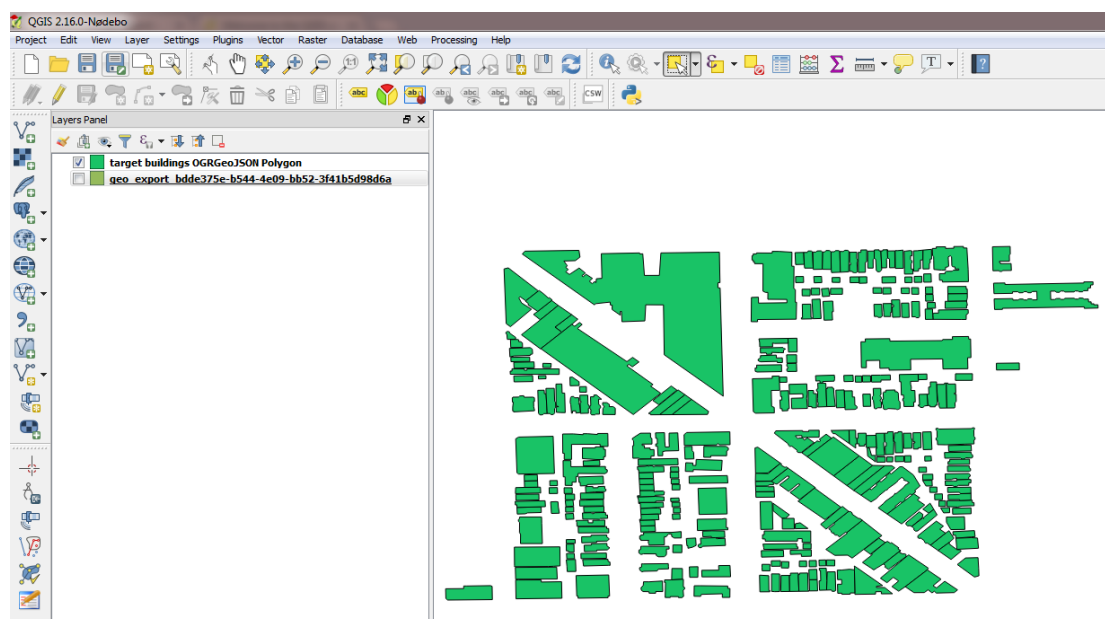


Figure 6. The target buildings in a new shapefile in QGIS

For the Building Retrofit analysis, the data are organized at the individual building level. For each building, the data in the following categories are required.

2.2. Building Basic Properties

Building basic properties data can be prepared in shapefile/GeoJSON format, or CSV/Excel file format, with one row for each building. For each row, you need to include the following properties for a building:

- ID (unique, linked to the building footprint file)
- Use type (building type)
- Year built
- Number of stories (above-grade)
- Building height (in meters)
- *Building Name (optional)*

The **use type (building type)** and **year built** are required to generate smart defaults for the building model, including some geometry parameters and system efficiencies. Typical open data sources for building type are zoning district maps and tax assessor records. Year built data can often be found in the tax assessor records or approximated from land annexation maps. If available, year remodeled should be prioritized over year of construction. See Section 4 for how to join the attributes from spatial datasets. The codes (case insensitive) for building types currently recognized by CityBES are shown in Table 3. If your city uses different naming conventions for use codes, you must convert your codes to one of the codes below. Future iterations of CityBES will recognize additional use types (ex. residential, industrial, mixed use). Feel free to mark buildings with these use types, but be aware that for now, all codes not listed below will be treated as "Other."

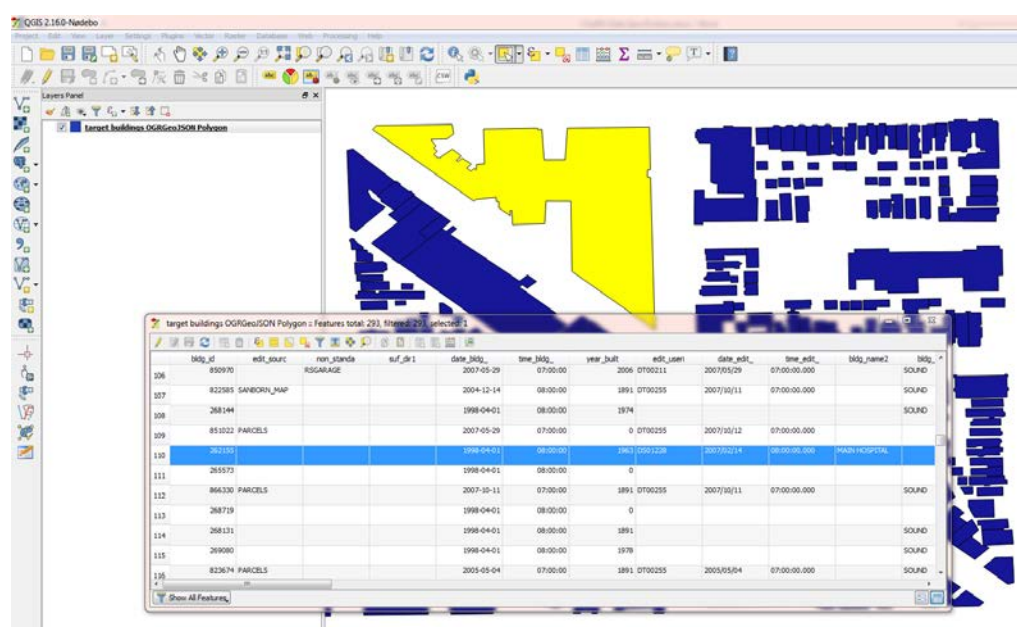
Table 3. Code list (case insensitive) for building types

Building Type	Code list (separated by comma)
Office	office, 211, 212, 500, b, bz, o, oa, oah, oal, oam, oat, obh, obl, obm, oc, och, ocl, ocm, omd, oz
Retail	retail, 202, 206, 208, 209, c, c1, cz, rh1, s
Restaurant	restaurant, 201
Hotel	hotel, h, h1, h2, hc, m, 110

The **number of stories** (above-grade) and **building height** are used together with the building footprint to generate the 3D geometry of the building model. If number of stories data are not available, estimate the number of stories based on the building height such that the floor-to-floor height is between 2.8-5.6m. The size of the building is calculated based on the building footprint and the number of stories. The **ID (unique)** provides the link to the building ID in the building footprint file. You can click the button to view the attributes of each building in QGIS. In this case, the **ID (unique)** should be the same as the **bldg_id** in Figure 7. If choosing to add the **building name**, make sure that the name does not consist solely of numbers. It is helpful to keep a user notes field to record any attribute assumptions and approximations you have made.

If you have the latitude and longitude information rather than the building ID, view Section 4 to join the building footprint with the point data.

If you have the address information only, view Section 5 to obtain longitude and latitude for each building and use Section 4 to complete the mapping.



The screenshot shows the QGIS 2.16.0-Nantes interface. The main map area displays a yellow building footprint on a dark blue background. The Layers Panel on the left shows 'target_buildings_OGRGeoJSON Polygon'. A table window titled 'target_buildings_OGRGeoJSON Polygon - Features total: 293, filtered: 293, selected: 3' is open, displaying the following data:

	bldg_id	est_sourc	est_stands	est_sur1	date_bldg	time_bldg	year_bldg	est_user	date_est	time_est	bldg_name2	bldg
106	850970		10540-A2		2007-05-29	07:00:00	2006	0700211	2007/05/29	07:00:00.000		SOUND
107	922885	SANBORN_MAP			2004-12-14	08:00:00	1891	0700255	2007/10/11	07:00:00.000		SOUND
108	268144				1998-04-01	08:00:00	1974					SOUND
109	851022	PARCELS			2007-05-29	07:00:00	0	0700255	2007/10/12	07:00:00.000		SOUND
110	262153				1998-04-01	08:00:00	1891	0700255	2007/10/14	08:00:00.000	MAIN 14 STPTN	
111	265573				1998-04-01	08:00:00	0					
112	866330	PARCELS			2007-10-11	07:00:00	1891	0700255	2007/10/11	07:00:00.000		SOUND
113	268719				1998-04-01	08:00:00	0					
114	268131				1998-04-01	08:00:00	1891					SOUND
115	269080				1998-04-01	08:00:00	1978					SOUND
116	823674	PARCELS			2005-03-04	07:00:00	1891	0700255	2005/05/04	07:00:00.000		SOUND

Figure 7. The building attributes in QGIS

2.3. Merge the Building Footprint with the Building Basic Properties

After creating the building basic properties, you need to merge the building basic properties with the building footprint. Figure 8 shows the function in QGIS to load the CSV file. In Figure 9, you need to right click on the building footprint dataset and select **Properties**. In the **Properties** popup window, select **Join** tab and click the **Add** button. From the **Add Vector Join** window, select the CSV dataset as Join Layer, and choose the associated building ID to join the dataset. Remember to check the **“Custom field name prefix”** and set it to empty unless you need to have a prefix. (Note: By default, QGIS use the file name as the “prefix”, which makes the filed name very long.) Save the **GeoJSON** file for CityBES.

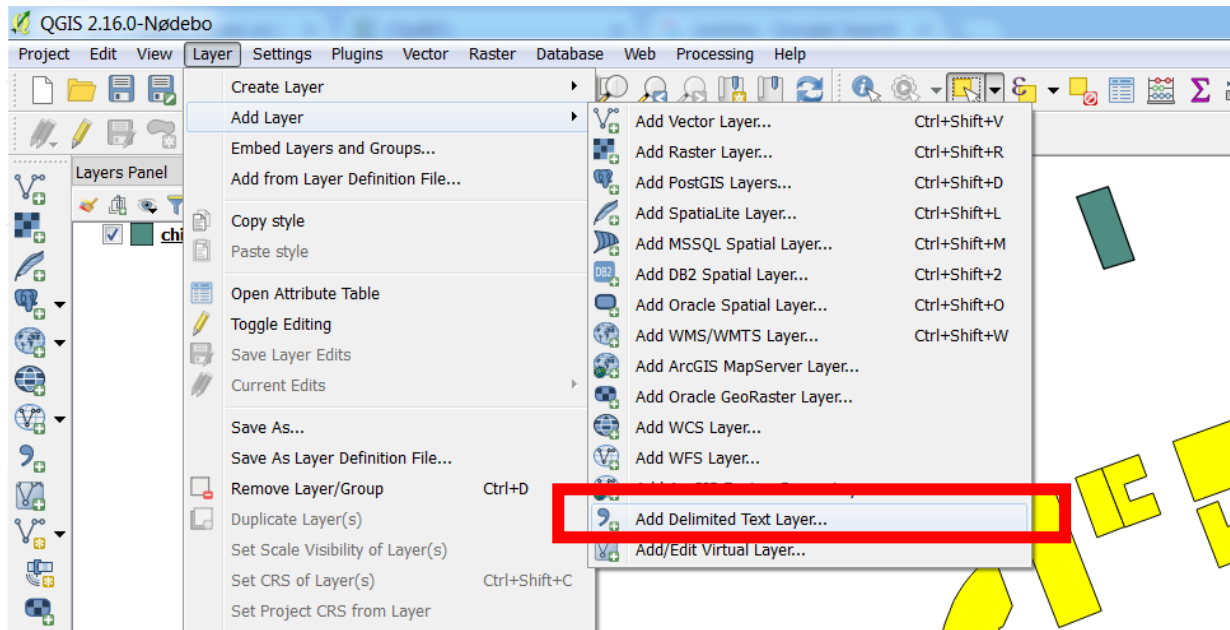


Figure 8. Load the building basic properties in csv file into QGIS

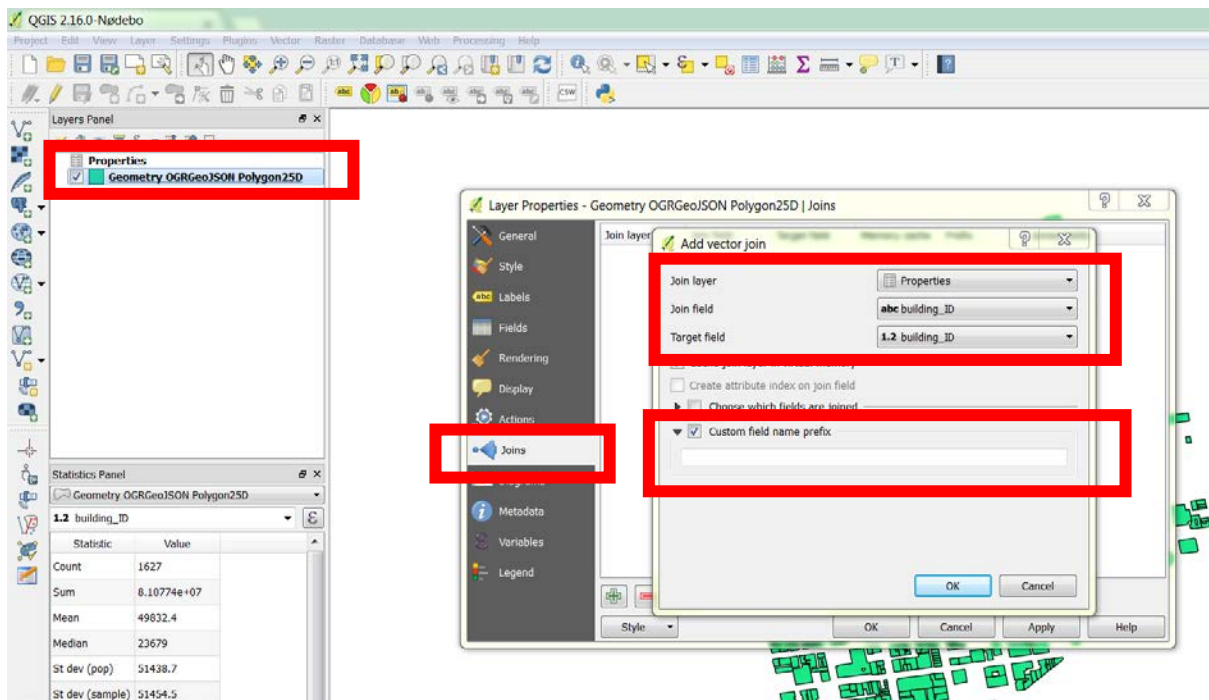


Figure 9. Join the building basic properties with building footprints

3. Optional Data for the CityBES Dataset

3.1. Building Energy Consumption Data

Building energy consumption data is used for energy benchmarking (e.g., obtaining an EnergyStar score),

and model calibration for the retrofit analysis. These data can be provided in CSV or Excel files with one row for each building. For each row, the following data can be included:

Title	Description	Unit / Choices
ID	The unique building ID	
eb_17_st	Benchmark status of 2017	Complied; Exempt; Violation - Did Not Report; Violation - Insufficient Data
ESScore_17	The energy star score of 2017	
SiteEUI_17	The site energy use intensity (EUI) of 2017	kBtu/(ft ² .year)
SorEUI_17	The source EUI of 2017	kBtu/(ft ² .year)
GHG_17	The GHG emission intensity of 2017	kg.CO ₂ e/ft ²
Elec_Y17	The annual electricity use of 2017	kWh
Gas_Y17	The annual natural gas use of 2017	Therm
Elec_Y17M01	The monthly electricity use of January 2017	kWh
Gas_Y17M01	The monthly natural gas use of January 2017	Therm

Change the 17 in the field title to 11, 12, 13, 14, 15 or 16 for years 2011, 2012, 2013, 2014, 2015, and 2016.

The **ID (unique)** provides the link to the building ID in the shapefile. You can click the  button to view the attributes of each building in QGIS. In this case, the **ID (unique)** should be the same as the **bldg_id** in the GeoJSON (Figure 7).

4. Join Polygon data (building footprints) and Point data (building properties) using QGIS

This section demonstrates the usage of QGIS to join polygon data (building footprints) and point data (csv data with longitude and latitude for each row).

Step 1: Load the Polygon data into QGIS

Load the building footprint into QGIS (Figure 10). Right click on the **Polygon dataset (Building Footprint)** and select **Properties**. In the **Properties** popup window, select **General** tab and change the **Coordinate reference system** to “**EPSG: 4326, WGS 84**” (Figure 12).

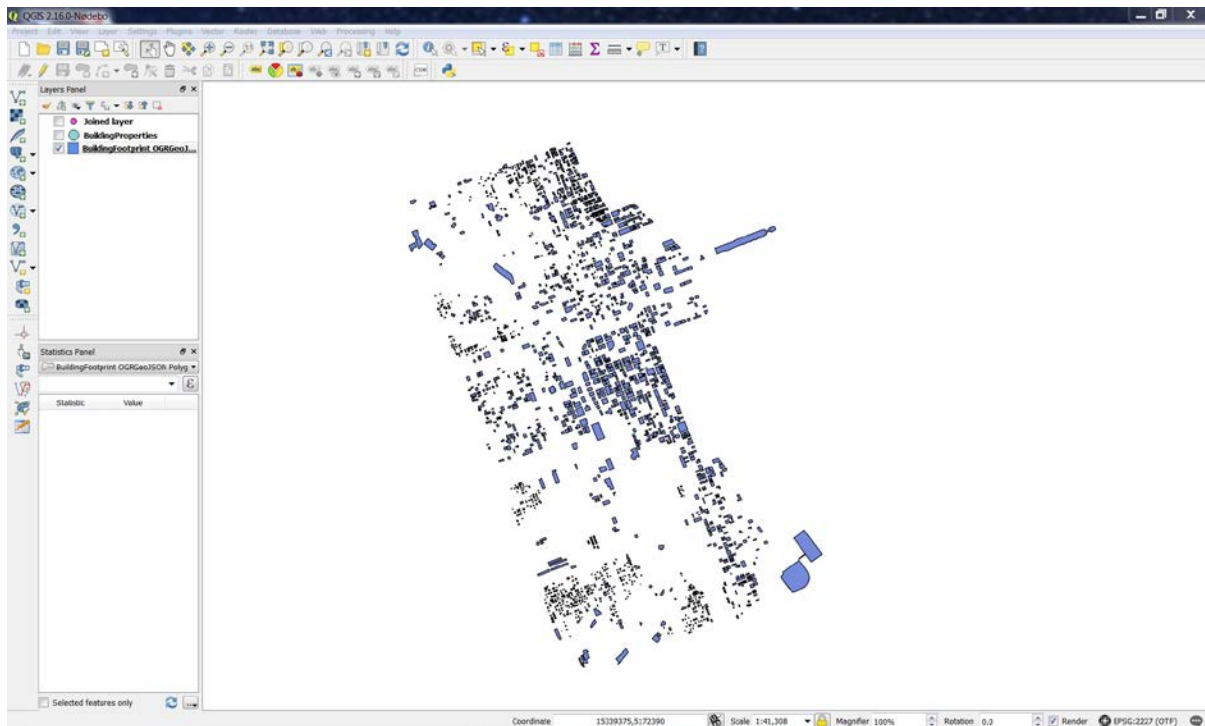


Figure 10. Load the building footprint into QGIS

Step 2: Load the Building Attributes into QGIS

Figure 8 shows the function in QGIS to load the CSV file. Figure 11 shows the setting to import a point data into QGIS. You need to select the **“Point Coordinates”** from the **“Geometry definition”**, and specify the X field (Longitude) and Y field (Latitude). If you don’t see the **Point** data overlap with the **Polygon** data, you need to change the **Coordinate Reference System** of the Point data. Right click on the **Point dataset (Building Properties)** and select **Properties**. In the **Properties** popup window, select **General** tab and change the **Coordinate reference system** to **“EPSG: 4326, WGS 84”** (Figure 12). You should see something similar to Figure 13 with the polygons overlap with the points.

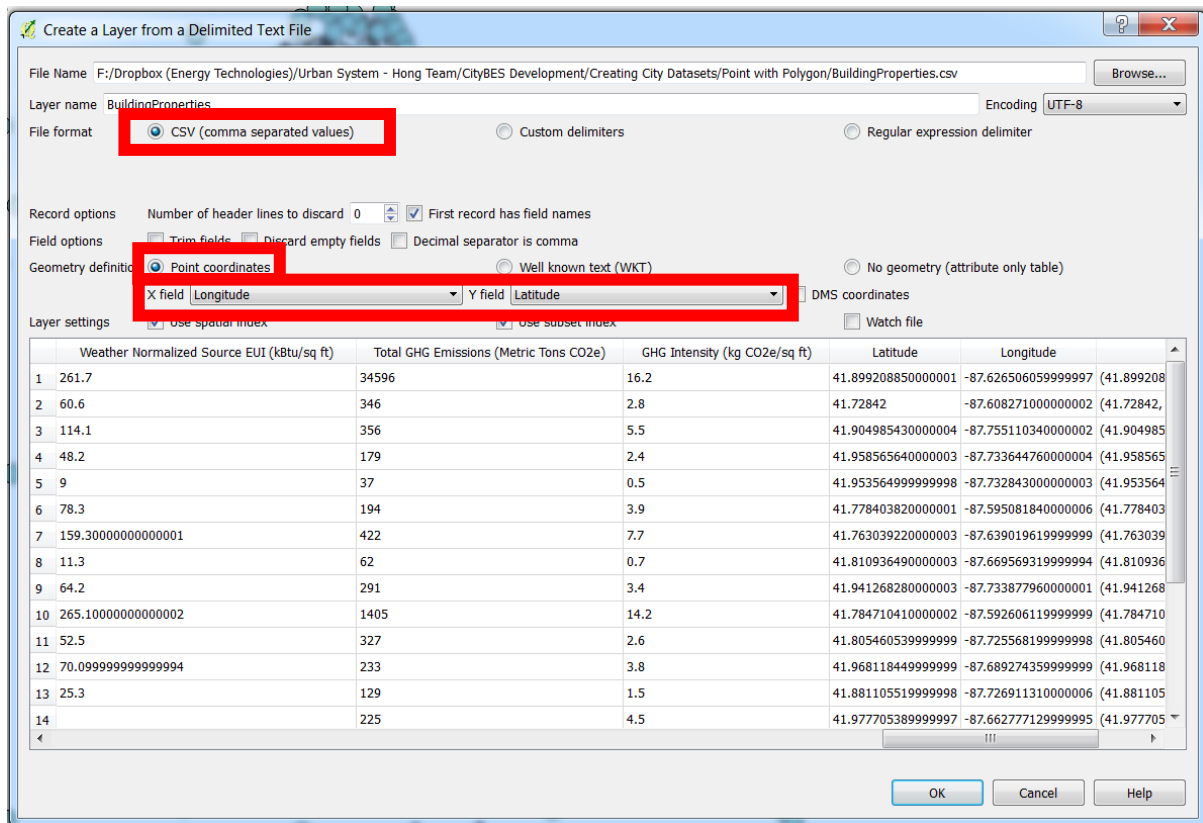


Figure 11. Load the building properties in CSV file into QGIS

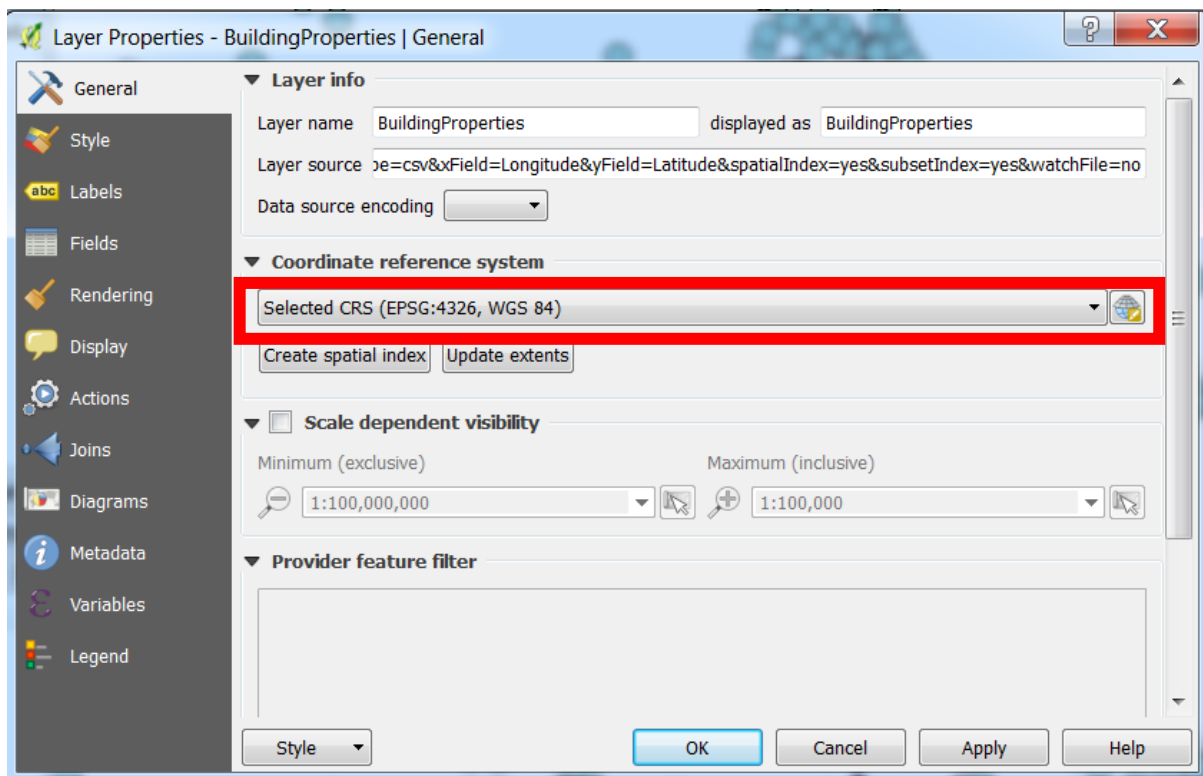


Figure 12. Update Coordinate reference system for building properties (points) in QGIS

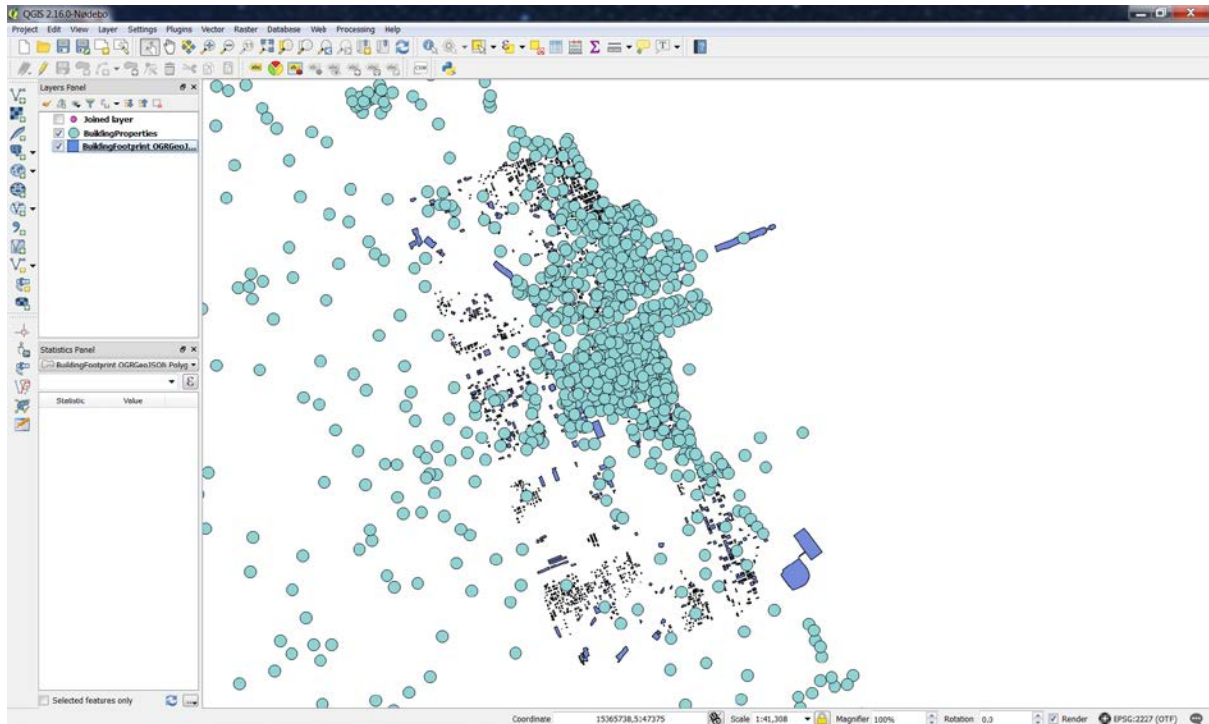


Figure 13. Building footprints (polygons) and building properties (points) in QGIS

Step 3: Join attributes by location

Follow Figure 14 to activate the **Join attributes by location** command. Figure 15 shows the setting to join attributes by location. The **Target vector layer** is the Point data (building properties), while the **Join vector layer** is the Polygon data (building footprint). For the **Geometric predicate**, select “**within**”. Figure 16 shows the results of joining attributes by location in QGIS. The results includes all the **points** that within a polygon. The **attributes** of the generated dataset (**Joined layer**) include the attributes from both the building footprint dataset and the building properties dataset. You should have the **unique building ID** in the building footprint dataset, which can be used to connect back to the building footprint.

If you switch data layers in the **Join attributes by location** as shown in Figure 17, the results will include all the **polygons** that contains a point (Figure 18).



Figure 14. Join attributes by location command in QGIS

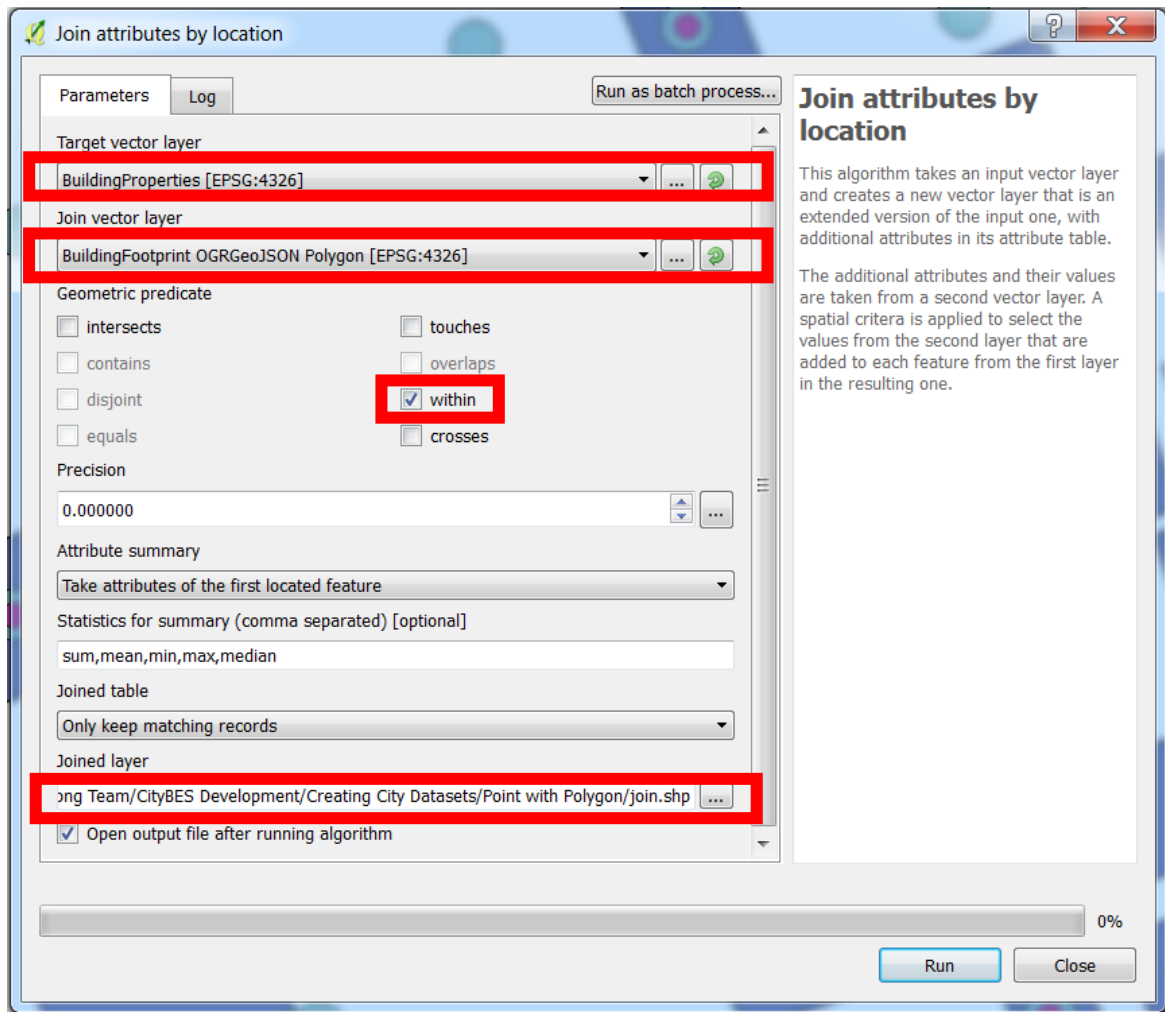


Figure 15. Join attributes by location window in QGIS

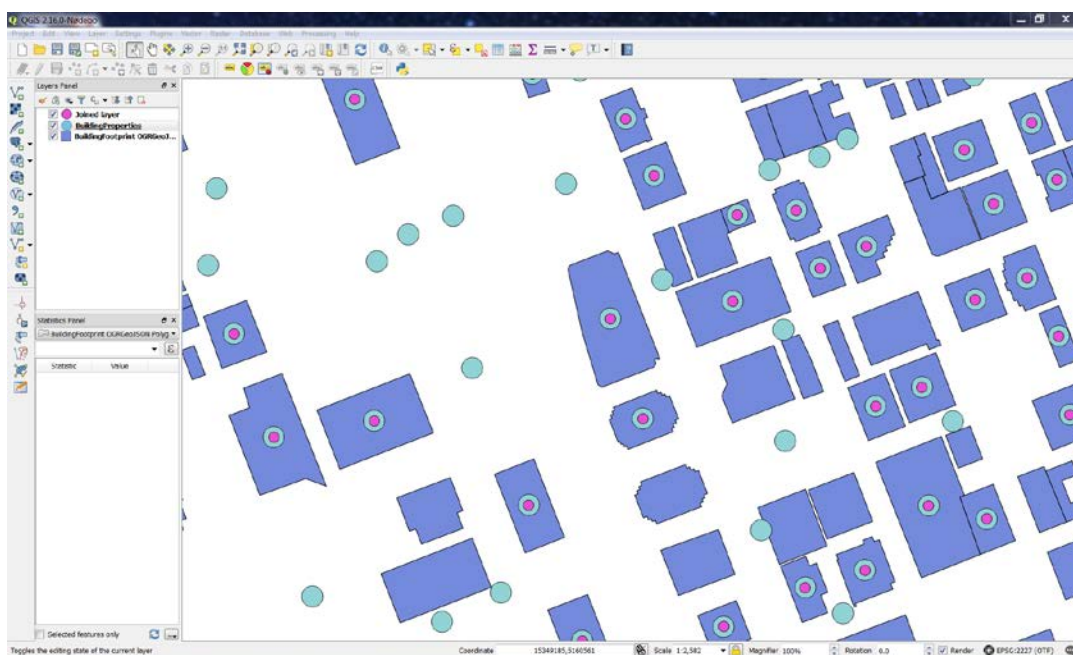


Figure 16. Results of joining attributes by location in QGIS

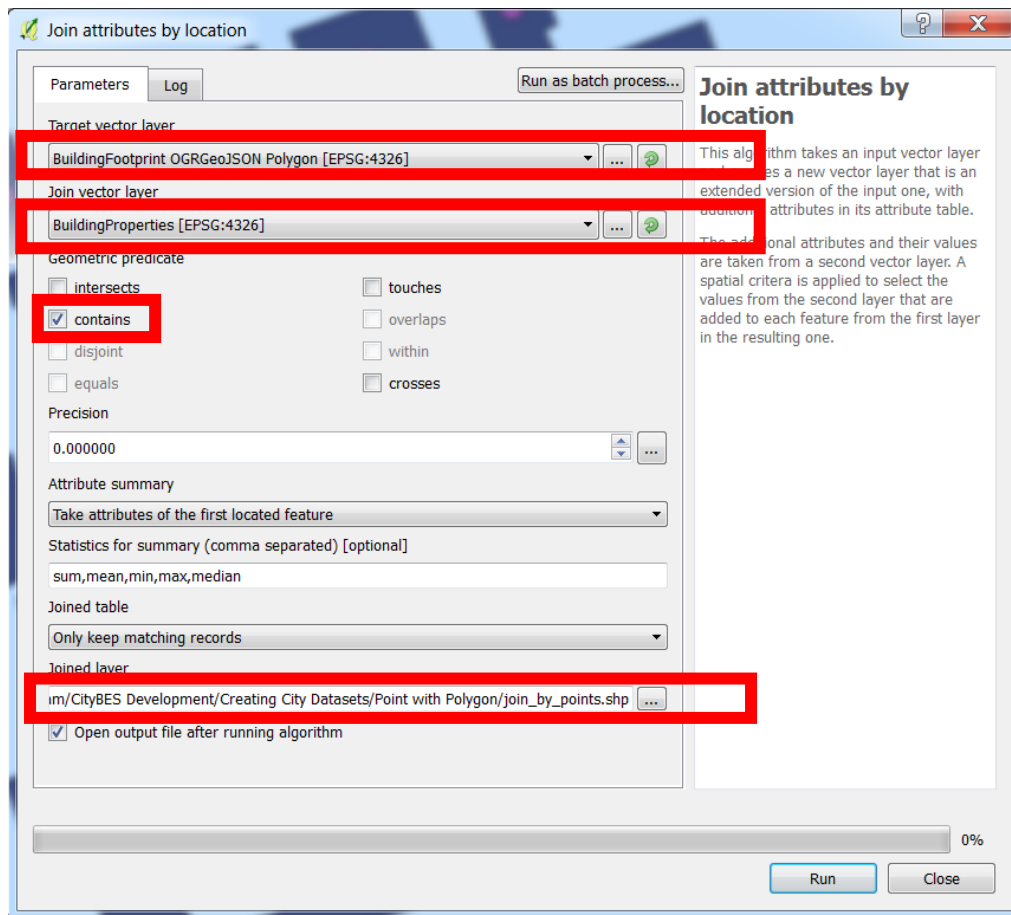


Figure 17. Switch data layers in QGIS

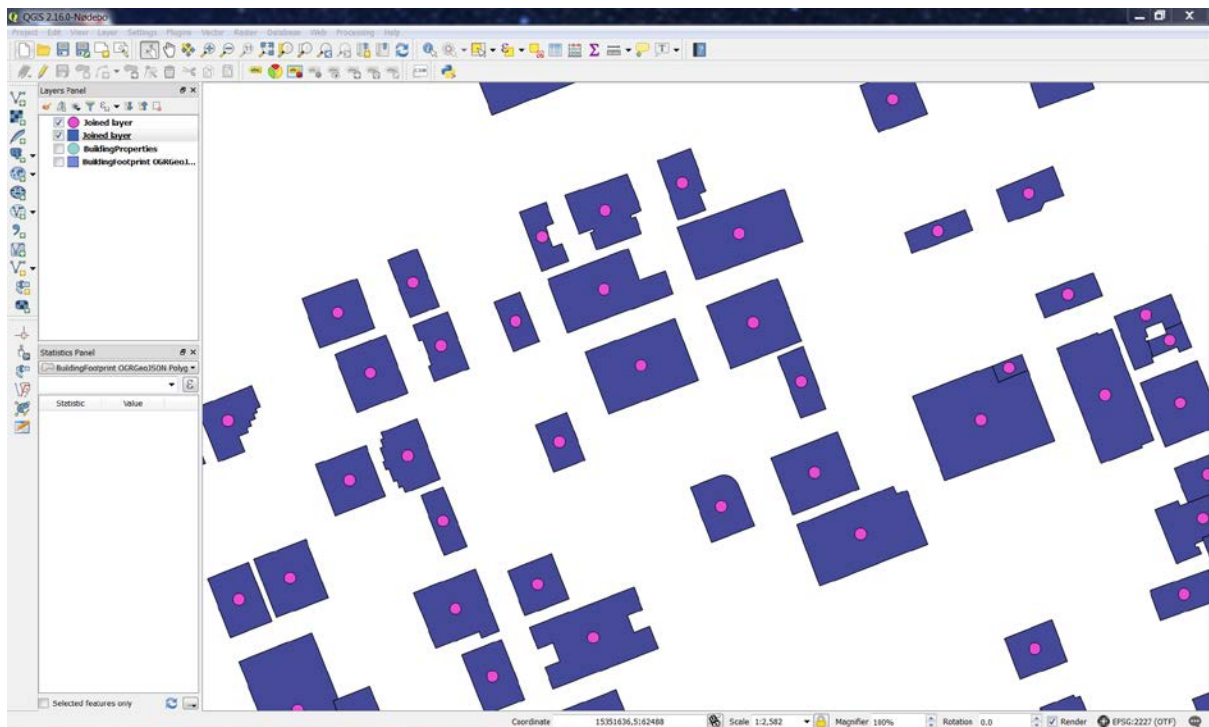


Figure 18. Results of joining attributes by location in QGIS

Step 4: Adjust the “Precision”

Sometimes, one point may be close to one polygon, but not inside the polygon. To address this issue, you can increase the “Precision” in Figure 15 or Figure 17. The precision value is the “longitude and latitude” distance rather than the value in feet or meet. So it may be in an order of 0.0001 rather than 1 or 100. You can use Google Map to find two points in your dataset, and compute their distance in term of longitude and latitude.

5. Obtain longitude and latitude information based on building address

This section introduces the resource to help obtain the longitude and latitude based on building’s address. A ruby script is developed to perform the task by query Google Geocode APIs. The Google Maps Geocoding API (<https://developers.google.com/maps/documentation/geocoding/start>) is a service that provides geocoding and reverse geocoding of addresses. You need to obtain a key from Google to use the script. Google provides 2500 free calls daily.

For detailed information, please look at the ruby script file <[citybes_obtain_lat_long.rb](#)>.

The ruby script file should be provided to you with this documentation. If you don’t find the ruby script file, please contact the development team for help.

- The encoding problem:
- Show errors when data load problem: 0/100 buildings error included
- The coordinate system in QGIS, do not change it in the properties

6. Split city building footprint data by planning district

CityBES currently supports datasets up to about 50 MB (roughly 25,000 buildings). To model an entire city, you must split the full dataset into smaller subdivisions, such as planning districts. To split the city building footprint by planning district, you will need the shapefiles/GeoJSON files for the full city building footprint and the planning districts. Planning district shapefiles are often available on a city's open data portal.

First, load both the building footprint and planning district boundary files into QGIS (Figure 19 below).

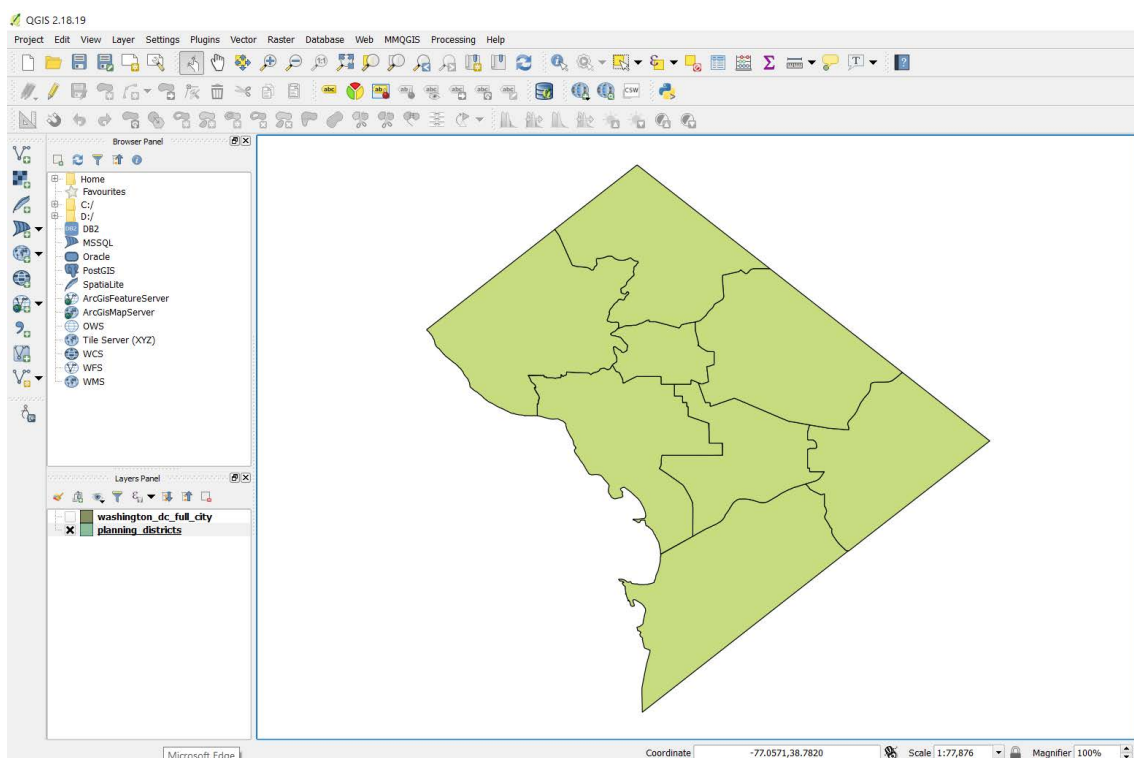


Figure 19. Shapefile of Planning District Boundaries

Then, use the split function to separate the single planning district shapefile into separate shapefiles for each district (see Figures 20 and 21 below).

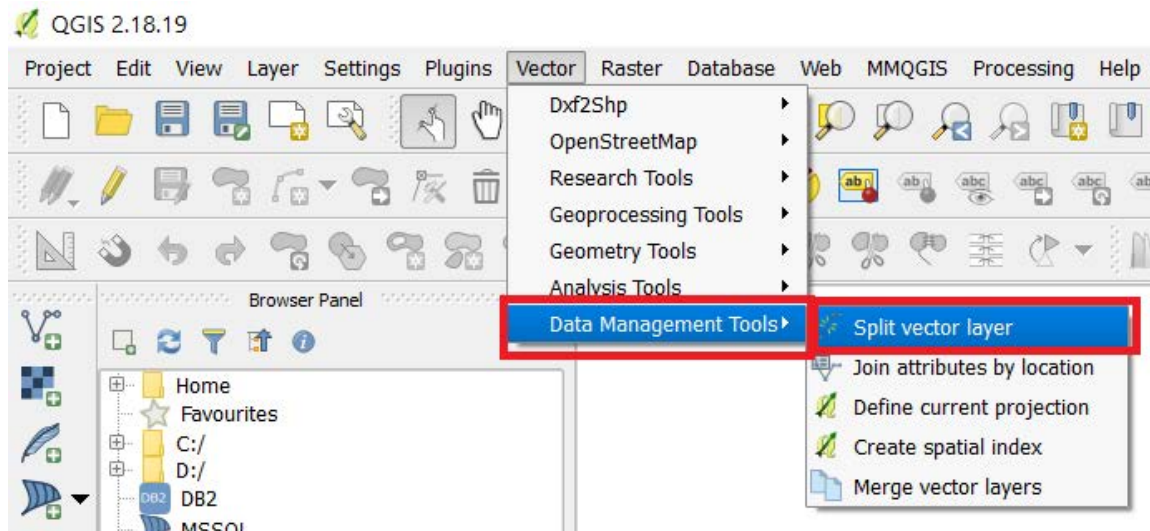


Figure 20: Split Vector Layer in QGIS

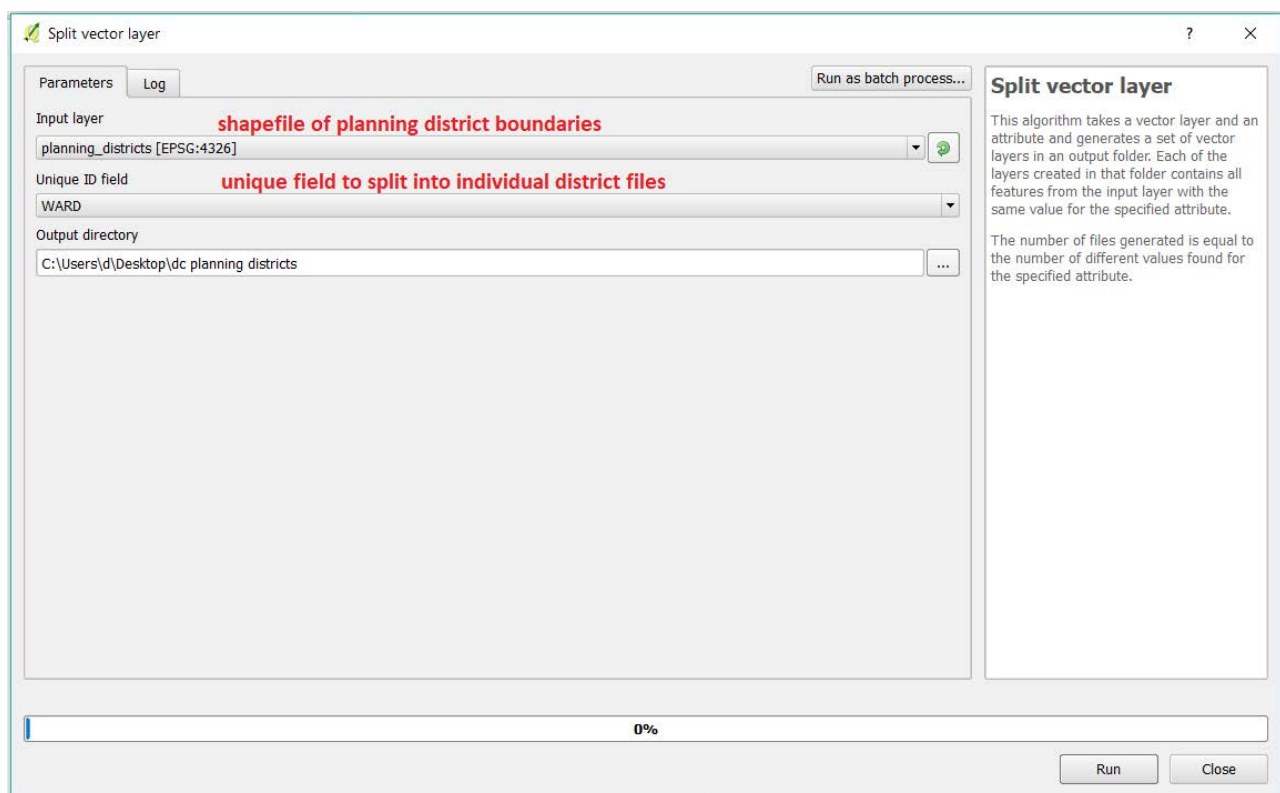


Figure 21: Splitting Single District Boundary Shapefile into Individual District Shapefiles

The individual district shapefiles can then be used to split the full city building footprint using the **clip function** (see Figures 22-24 below).

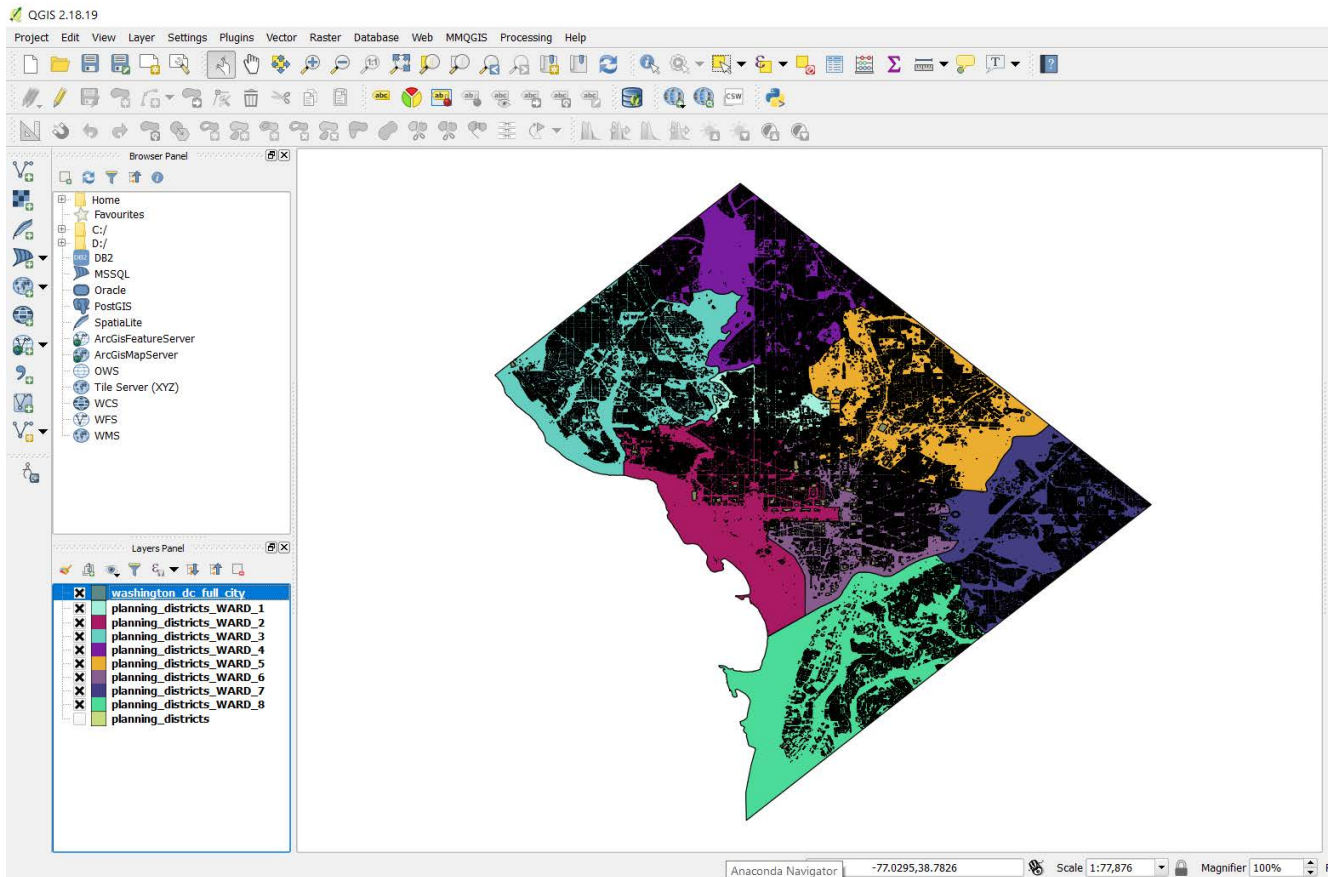


Figure 22: City Building Footprint Overlaying Individual Planning District Shapefiles

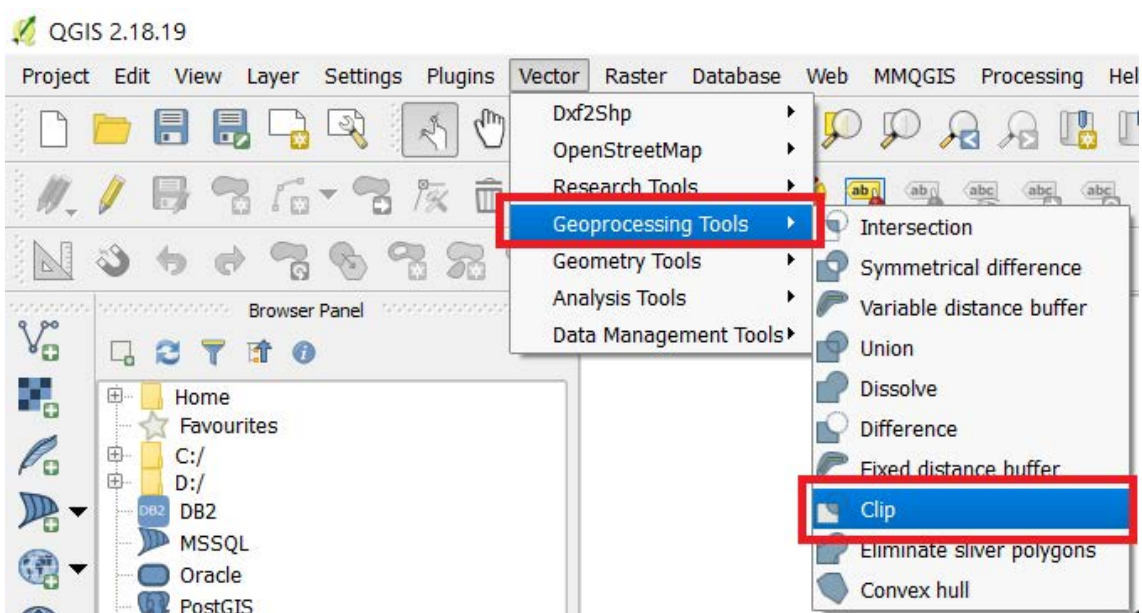


Figure 23: Clip Vector Layer in QGIS

Clip

ParametersLogRun as batch process...

Input layer

planning_districts [EPSG:4326]

Clip

This algorithm clips a vector layer using the polygons of an additional polygons layer. Only the parts of the features in the input layer that falls within the polygons of

Batch Processing - Clip

ParametersLog

+

+

+

+

Input layer		Clip layer		Clipped	
washington_dc_full_city	...	planning_districts_WARD_1	...	op/dc planning districts/footprint_district_1.geojson	...
washington_dc_full_city	...	planning_districts_WARD_2	...	op/dc planning districts/footprint_district_2.geojson	...
washington_dc_full_city	...	planning_districts_WARD_3	...	op/dc planning districts/footprint_district_3.geojson	...
washington_dc_full_city	...	planning_districts_WARD_4	...	op/dc planning districts/footprint_district_4.geojson	...
washington_dc_full_city	...	planning_districts_WARD_5	...	op/dc planning districts/footprint_district_5.geojson	...
washington_dc_full_city	...	planning_districts_WARD_6	...	op/dc planning districts/footprint_district_6.geojson	...
washington_dc_full_city	...	planning_districts_WARD_7	...	op/dc planning districts/footprint_district_7.geojson	...
washington_dc_full_city	...	planning_districts_WARD_8	...	op/dc planning districts/footprint_district_8.geojson	...

Full City Building Footprint

Planning District Boundary Shapefiles

Building Footprint by District

0%

Figure 24: Clip City Building Footprint by Planning District